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DESIGN AND SIMULATION OF TEMPERATURE SENSOR CIRCUIT

AIM:

To design and simulate a temperature sensor circuit and display the output on an LCD and create a PCB for the circuit.

SOFTWARE REQUIRED:

- KiCAD or Altium Designer.

COMPONENTS REQUIRED:

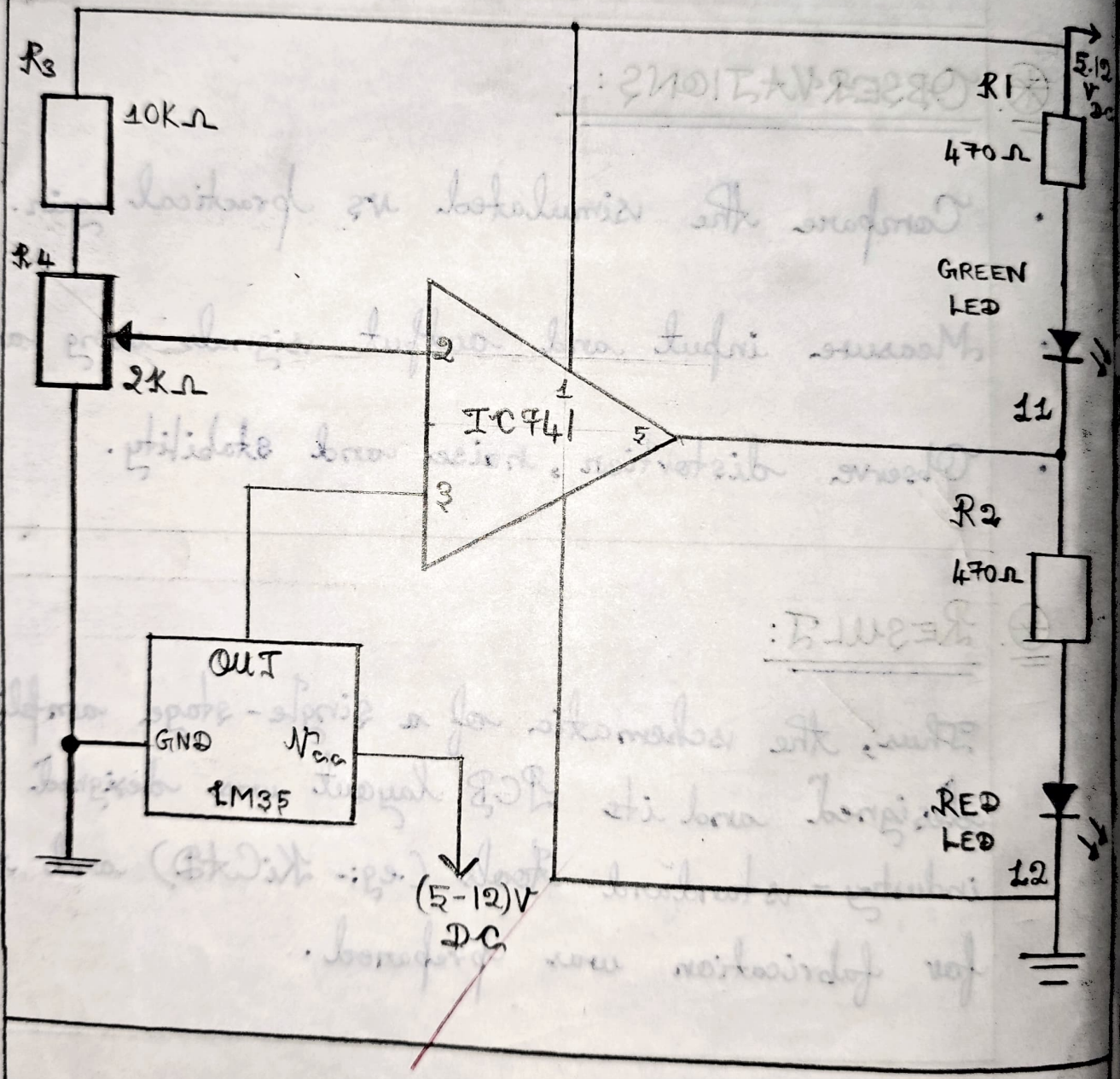
- Temperature Sensor - LM35.
- Operational Amplifier - IC 741.
- Resistors - 470 Ω, 10 KΩ, 2 KΩ.
- PCB Board.
- Power Supply (5V-12V).
- LEDs.
- Microcontroller.

CIRCUIT DESIGN:

1. Sensor Connection:

- Connect V_{cc} to 5V (or 3.3V).
- Connect GND to Ground.
- Connect the VOUT of LM35 to an ADC pin of the microcontroller.

CIRCUIT DIAGRAM:



2. Signal Conditioning (if needed):

- Use an Op-Amp (IC 741) as a buffer to stabilize the signal.
- A low-pass filter (RC circuit) can be added to remove noise.

3. Microcontroller Processing:

Read the analog voltage from the sensor.

Convert to temperature using:

$$T(^\circ C) = \frac{V_{out}\,(mV)}{10}$$

- Display the temperature on an LCD or Serial Monitor.

⊛. PCB LAYOUT DESIGN:

1. Software Tools:

- Proteus (for circuit simulation).
- KiCAD/Eagle/Altium (for PCB design).

2. Steps:

- Place components logically.
- Route the power and ground planes efficiently.
- Ensure thermal relief pads for the sensor.
- Minimize noise by keeping analog and digital sounds separate.

- Generate Gerber files for PCB fabrication.

PROCEDURE :

1. **Circuit Testing on Bread board:**

- Assemble the circuit and verify temperature readings using a multimeter and an oscilloscope.

2. **Simulation in Proteus or LTSpice:**

- Validate circuit performance before PCB fabrication.

3. **PCB Fabrication:**

- Etch or use PCB manufacturers for a professional layout.
- Solder components onto the PCB.

4. **Final Testing:**

- Verify the output voltage against a reference thermometer.
- Check PCB performance for accuracy, noise, and reliability.

OBSERVATIONS AND ANALYSIS:

- Compare sensor output voltage vs actual temperature.
- Identify errors and possible improvements in the PCB design.
- Discuss the effect of noise, power supply variation, and PCB layout on sensor performance.

RESULT:

Thus, the temperature sensor circuit to display the output on an LCD was designed and simulated and the PCB for the circuit was created.